

"Embedded software implementations are written in 'C' language and executed on digital signal processors or microcontrollers. Hardware implementations are written in hardware description languages (HDLs) such as VHDL or Verilog and programmed onto FPGA or realised on ASICs. Being able to reuse simulation models for implementing algorithms in 'C' or HDL languages by employing automatic code generation technologies like Embedded Coder and HDL Coder enables design teams to leverage the effort invested in creating simulation models, reduces the overall product development time, and saves expensive and lengthy design iterations by mitigating coding and interpretation errors," explains Prashant Rao, technical manager, application engineering & training, MathWorks India.

There are two big advantages to doing a simulation before actually building the design and testing it.

"The biggest of these advantages is money. Designing, building, testing, redesigning, rebuilding and retesting for anything can be an expensive project in itself. Simulations take the building/rebuilding phase out of the loop by using the model already created in the design phase. Most of the time simulation testing is cheaper and faster than performing multiple tests of the design each time," explains T. Anand, managing director, Knewron.

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"Apart from cost savings, the second biggest advantage of simulation is the level of detail that you can get from simulation. Simulation can give you results that may not be experimentally measurable with our current level of technology or available instruments in hand" — T. Anand, MD, Knewron

set the simulation to run for as many time steps as you desire and at any level of detail. The only restrictions are your imagination, programming skills, etc."

Besides, design firms benefit from time-to-market reduction and savings on expensive resources. Since many tests may call for destructive mechanism, costly components cannot be used frequently. Simulation helps to save a lot on that part. It also saves a lot of time in re-iterations being done physically and thus improves time-to-market.

Addressing system complexity

With system complexity increasing rapidly, and in some cases even exponentially, simulation models that capture and mirror the behaviour of systems have also become extremely large.

"Simulation models are increasingly being used to communicate and share design specifications between project partners or between original equipment manufacturers (OEMs) and suppliers. Large corporations are using model-based design across product development processes spanning development and application centres across the globe. To collaborate, they have really large models—some with

millions of blocks that are created or assembled by hundreds of engineers. A simulation environment has to be able to handle huge models that are developed by hundreds of engineers from diverse domains across geographically dispersed markets," explains Rao.

The new MATLAB Desktop features a Toolstrip that puts key features and commonly-used functionalities upfront, a gallery of pre-built MATLAB apps, and tools that enable customers to create and share apps of their own. Simulink platform for model-based designs now features the capability to step through as well as rewind a simulation. Stateflow also incorporates the capability to create state-transition tables and use MATLAB as the Action Language.

As far as energy is concerned, virtually every organisation is facing challenges to make equipment that are more energy-efficient than their predecessor.

"It should take up less energy and do more work. I remember talking to a key executive from a product manufacturing company, who said that today he has to build a product that has half the size and twice the rating of those made a mere five years ago. That's the kind of challenges energy and rotor machinery manufacturers and product developers are facing," explains Dutta.

The industry has grown such that an engineer might have to work with a multi-disciplinary approach to successfully deliver cutting-edge products. Simulation tools now feature add-on modules that enable engineers to speed things up.

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